

Skin Cancer Detection through Deep Learning using Skin Lesion Images

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Department of Artificial Intelligence and Data Science

Course: Deep Learning (UG)

Academic Year: 2025–2026 | Semester: IV

Reference Paper DOI: 10.1038/s41598-025-26392-4

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Abstract

Skin cancer is among the most prevalent and life-threatening malignancies worldwide, with global incidence rates rising by 33% between 2007 and 2017. Early and accurate detection is critical for improving patient survival outcomes. However, conventional diagnosis relies heavily on subjective expert visual inspection, which is time-consuming and prone to inter-observer variability. This project presents an automated deep learning system for multi-class skin lesion classification using dermoscopic images from the ISIC (International Skin Imaging Collaboration) dataset, covering nine clinically distinct lesion categories.

The proposed methodology employs EfficientNet-B2, a compound-scaled convolutional neural network pretrained on ImageNet, as the primary classification model. Transfer learning is applied by fine-tuning the pretrained weights on the target dataset, enabling effective feature extraction from limited medical image data. A custom Baseline CNN is also developed from scratch as a reference model to quantitatively demonstrate the performance gains achieved through transfer learning.

To address class imbalance inherent in the dataset, WeightedRandomSampler and class-weighted CrossEntropyLoss are employed during training. The model is optimized using AdamW with CosineAnnealingLR scheduling over 30 epochs. Comprehensive evaluation is conducted using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC analysis. Additionally, Grad-CAM (Gradient-weighted Class Activation Mapping) is integrated to generate visual heatmap explanations, highlighting lesion regions

influencing model predictions and supporting clinical interpretability. Experimental results demonstrate that EfficientNet-B2 with transfer learning significantly outperforms the custom Baseline CNN, validating the effectiveness of the proposed approach for automated skin cancer screening.

Keywords — Skin cancer detection, deep learning, EfficientNet-B2, transfer learning, ISIC dataset, multi-class classification, Grad-CAM, class imbalance, dermoscopy, medical image analysis.

I. Introduction

Skin cancer is one of the fastest-growing cancers globally, with melanoma alone projected to reach 510,000 new cases and approximately 96,000 deaths annually by 2040 [1]. The global shortage of dermatology specialists and increasing patient volume exacerbate delays in diagnosis, adversely impacting treatment outcomes. Traditional clinical diagnosis relies on expert visual inspection of dermoscopic images, a process that is inherently subjective, time-intensive, and subject to inter-observer variability.

Deep learning has demonstrated near-dermatologist-level performance in medical image classification tasks [2]. Convolutional neural networks (CNNs) can automatically learn hierarchical feature representations directly from raw image data, eliminating the need for hand-engineered features. Transfer learning, which leverages representations learned from large-scale datasets such as ImageNet, has further enabled high-performing models to be trained on comparatively small medical imaging datasets.

Despite promising progress, several challenges remain unresolved in the literature: the ability to simultaneously classify multiple lesion types, handling severe class imbalance in clinical datasets, and the critical requirement for model interpretability in healthcare settings. Black-box deep learning models are difficult to adopt clinically without explanations for their predictions. Gradient-weighted Class Activation Mapping (Grad-CAM) addresses this by generating visual explanations that highlight the image regions driving model decisions.

This project addresses these challenges by implementing an end-to-end automated skin lesion classification system using EfficientNet-B2 on the ISIC dataset across nine lesion categories, with integrated Grad-CAM explainability and a baseline CNN comparison.

II. Literature Survey

A comprehensive review of existing works in deep learning-based skin cancer detection reveals substantial progress alongside persistent limitations. Table I summarizes ten significant contributions drawn from the literature.

TABLE I — Literature Survey Summary

No .	Authors & Year	Model / Approach	Dataset	Limitation
1	Esteva et al. (2017)	InceptionV3 + Transfer Learning	129K clinical images	Binary classification ; limited explainability
2	Brinker et al. (2019)	ResNet50 Fine-Tuned CNN	ISIC 2016	Focused on melanoma vs. nevus only
3	Tan & Le (2019)	EfficientNet	ImageNet	General-purpose; not

No .	Authors & Year	Model / Approach	Dataset	Limitation
		(Compound Scaling)		medical-specific
4	Chaturvedi et al. (2019)	MobileNet Transfer Learning	HAM10000	Moderate multiclass performance
5	Lee et al. (2018)	Fine-Tuned Neural Network Ensemble	ISIC 2018	High computational complexity
6	Venugopal et al. (2023)	Modified EfficientNet + DNN	Combined dermoscopic	Binary emphasis; training complexity
7	Naeem et al. (2024)	CNN + Handcrafted Feature Fusion	ISIC 2019	High feature engineering overhead
8	Kanchana et al. (2024)	EfficientNet B0–B7 Transfer Learning	Skin lesion dataset	Sensitive to preprocessing pipeline
9	Sandag et al. (2025)	EfficientNet + Grad-CAM	Kaggle dermoscopic	Limited to binary melanoma classification
10	Islam et al. (2026)	EfficientNet-B2 + Metadata Fusion	79,246 UK patient images	Proprietary dataset; limited reproducibility

Esteva et al. [2] demonstrated dermatologist-level performance using InceptionV3 on 129,000 clinical images; however, the study was confined to binary classification, and the model lacked explainability mechanisms. Brinker et al. [3] extended this to a benchmark study comparing ResNet50 against 157 dermatologists on ISIC 2016 data, but again focused narrowly on melanoma versus nevus discrimination. Tan and Le [4] introduced the EfficientNet family with compound scaling, establishing state-of-the-art performance on ImageNet, though without domain-specific adaptation to medical imaging.

Subsequent works explored lightweight architectures (MobileNet) [4], ensemble strategies

[5], and hybrid CNN-feature fusion approaches [7]. Kanchana et al. [8] evaluated EfficientNet variants B0 through B7 for skin lesion classification but noted sensitivity to preprocessing pipeline design. Most recently, Sandag et al. [9] applied EfficientNet with Grad-CAM but restricted evaluation to binary melanoma classification. Islam et al. [10], the primary reference for this project, demonstrated EfficientNet-B2 with patient metadata fusion achieving superior performance on a large proprietary UK patient dataset. The common limitations across these works — binary classification scope, class imbalance neglect, and lack of explainability — directly motivate the contributions of the present study.

III. Research Gaps

Based on the reviewed literature, four primary research gaps are identified. Table II presents these gaps alongside the specific contributions of the present work.

TABLE II — Research Gaps and Proposed Contributions

Research Gap Identified	Our Contribution
Class Imbalance in Medical Datasets	WeightedRandomSampler + Class-weighted CrossEntropyLoss to ensure balanced learning across all 9 classes
Lack of Explainability in CNN Models	Grad-CAM heatmap visualization applied to the final convolutional layer of EfficientNet-B2
Overfitting on Small Medical Datasets	Transfer learning (ImageNet pretrained weights) + data augmentation + AdamW weight decay regularization

Research Gap Identified	Our Contribution
Binary-Only Classification in Prior Works	9-class simultaneous multi-class classification covering the full ISIC lesion taxonomy

Class imbalance is a fundamental challenge in clinical datasets, where rare lesion types are significantly underrepresented relative to common ones. Prior works often overlook this issue or rely on simple oversampling. The absence of explainability in deployed CNN models constitutes a critical barrier to clinical trust and regulatory approval. Overfitting on small medical datasets is a recurring problem that transfer learning and augmentation can substantially mitigate. Finally, the restriction of most prior works to binary classification limits their direct clinical applicability, as dermatologists routinely differentiate among multiple lesion categories.

IV. Objectives

The objectives of this project are:

- To develop a robust multi-class skin lesion classification system capable of identifying nine major categories of skin diseases using the ISIC dataset.
- To implement EfficientNet-B2 with ImageNet pretrained weights via transfer learning for efficient and accurate feature extraction on limited medical imaging data.
- To design an effective image preprocessing and data augmentation pipeline to improve model generalization.
- To address class imbalance using WeightedRandomSampler and class-

weighted loss functions to ensure balanced learning across all nine lesion categories.

- To comprehensively evaluate model performance using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC analysis.
- To integrate Grad-CAM-based explainability for generating clinically interpretable heatmap visualizations supporting trustworthy AI-assisted diagnosis.
- To establish a quantitative performance comparison between the proposed EfficientNet-B2 model and a custom Baseline CNN trained from scratch.

V. Dataset Description

The International Skin Imaging Collaboration (ISIC) dataset [10] is a widely adopted benchmark for dermoscopic image analysis. It comprises high-resolution dermoscopy images across nine clinically distinct skin lesion categories. The dataset was structured into dedicated training and test splits, with a further 85%/15% stratified subdivision of the training set into training and validation partitions.

TABLE III — ISIC Dataset Class Distribution

Lesion Class	Train	Test	Total
Actinic Keratosis	114	16	130
Basal Cell Carcinoma	376	16	392
Dermatofibroma	95	16	111
Melanoma	438	16	454
Nevus	357	16	373
Pigmented Benign Keratosis	462	16	478
Seborrheic Keratosis	77	3	80
Squamous Cell Carcinoma	181	16	197

Lesion Class	Train	Test	Total
Vascular Lesion	139	3	142
Total	2239	118	2357

The dataset exhibits notable class imbalance, with Pigmented Benign Keratosis (462 training images) and Melanoma (438 training images) comprising the largest categories, while Seborrheic Keratosis (77) and Dermatofibroma (95) represent the minority classes. This 6:1 imbalance ratio between the most and least frequent classes necessitates dedicated handling strategies during training. All images were resized to 224×224 pixels to conform with EfficientNet-B2's input requirements.

VI. Proposed Methodology

The proposed methodology consists of a sequential pipeline from raw data to interpretable predictions, as described in the following subsections.

A. Data Preprocessing

All input images are resized to 224×224 pixels and normalized using ImageNet channel statistics (mean = [0.485, 0.456, 0.406]; std = [0.229, 0.224, 0.225]). This normalization ensures that the pixel value distribution at inference is consistent with the distribution on which EfficientNet-B2's pretrained weights were computed, preserving the validity of transferred feature representations.

B. Data Augmentation

To improve generalization and reduce overfitting on limited medical data, the following augmentations are applied exclusively to training images:

- Random Horizontal Flip — simulates mirror-image lesion orientations.

- Random Vertical Flip — adds rotational diversity.
- Random Rotation ($\pm 20^\circ$) — accounts for varying dermoscope orientations during image acquisition.
- ColorJitter (brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1) — simulates variability in clinical imaging conditions and lighting environments.

Test images are processed exclusively with deterministic transforms (Resize and Normalize) to ensure evaluation reflects true generalization performance on unseen data.

C. Class Imbalance Handling

Two complementary strategies are employed to mitigate class imbalance. Weighted Random Sampler assigns each training sample a sampling probability inversely proportional to its class frequency, such that rare classes appear with equivalent frequency to majority classes within each mini-batch. Class-weighted Cross Entropy Loss multiplies the per-sample loss contribution by a class weight defined as:

$$weight(c) = N_{total} / (K \times N_c)$$

where N_{total} is the total number of training samples, K is the number of classes (9), and N_c is the count of class c . This formulation ensures that misclassification of minority class instances incurs proportionally greater gradient updates, directing the optimizer toward balanced performance.

D. Model Training

Both EfficientNet-B2 and the Baseline CNN are trained using the AdamW optimizer with a learning rate of 3×10^{-4} and weight decay of 1×10^{-4} . A CosineAnnealingLR scheduler reduces the learning rate smoothly from LR_{max} to 0 over $T_{max} = 30$ epochs, preventing abrupt learning

rate drops and enabling convergence to better local minima. The best model checkpoint (by validation accuracy) is saved at each epoch.

E. Evaluation and Explainability

Both models are evaluated on the held-out test set using accuracy, per-class precision, recall, F1-score, confusion matrix, and macro-average ROC-AUC. Grad-CAM is applied to the final convolutional layer of EfficientNet-B2 to generate spatial heatmaps that visualize which regions of each dermoscopic image contribute most to the predicted class. These visualizations serve as a clinical interpretability mechanism.

VII. EfficientNet-B2 Architecture

EfficientNet [4] is a family of CNN architectures developed by Tan and Le (Google Brain, 2019) that achieves superior accuracy-efficiency trade-offs through compound scaling. Rather than scaling network depth, width, or input resolution individually — as is common in prior architectures such as ResNet and VGG — EfficientNet uniformly scales all three dimensions simultaneously using a fixed compound coefficient ϕ :

depth: $d = \alpha^\phi$, *width:* $w = \beta^\phi$, *resolution:* $r = \gamma^\phi$
subject to the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, with constants α , β , γ determined by a neural architecture search on a baseline B0 model. This principled scaling strategy produces a family of eight models (B0–B7) with monotonically increasing capacity and accuracy.

MBConv Blocks: The fundamental building block of EfficientNet is the Mobile Inverted Bottleneck Convolution (MBConv) block, inherited from

MobileNetV2. Each MBConv block performs depthwise separable convolution — separating spatial filtering (depthwise) from channel mixing (pointwise) — drastically reducing parameters while preserving representational capacity. EfficientNet further augments MBConv blocks with Squeeze-and-Excitation (SE) attention, which adaptively recalibrates channel-wise feature responses based on global context, enabling the network to focus on the most informative feature channels.

Transfer Learning: EfficientNet-B2 is initialized with weights pretrained on ImageNet (1.28M images, 1000 classes). These weights encode rich hierarchical visual features — edge detectors in early layers, texture patterns in intermediate layers, and semantic object-part representations in deeper layers. For skin lesion classification, the final fully connected classification layer is replaced with a new linear layer mapping to 9 output classes, and the entire network is fine-tuned end-to-end on the ISIC training data. This approach is particularly effective when the target domain (dermoscopy) shares low- and mid-level visual statistics with the source domain (natural images).

Why EfficientNet-B2: Among the EfficientNet variants, B2 offers an optimal balance between classification accuracy and computational cost, making it suitable for training on a dataset of approximately 2,000 images without requiring high-performance GPU infrastructure. With approximately 9.1 million trainable parameters, it is significantly more expressive than the custom Baseline CNN (~2.6M parameters) while remaining tractable for the available hardware.

VIII. Baseline CNN Architecture

A custom Baseline CNN was designed from scratch to serve as a quantitative reference model. The architecture follows a standard VGG-like pattern of stacked convolutional blocks followed by a fully connected classifier:

Input $\rightarrow$ [Conv $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Conv $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Dropout] $\times 4 \rightarrow$ FC $\rightarrow$ Output

Each convolutional block progressively doubles the number of feature maps: Block 1 (32 channels), Block 2 (64 channels), Block 3 (128 channels), Block 4 (256 channels). MaxPooling with stride 2 halves the spatial dimensions after each block, reducing the 224×224 input to a 4×4 feature map before global average pooling. Batch Normalization is applied after each convolutional layer to stabilize training. Dropout rates increase progressively (0.10, 0.15, 0.20, 0.25) to provide stronger regularization in deeper blocks.

The classifier consists of a fully connected layer ($256 \times 4 \times 4 \rightarrow 512$), ReLU activation, Dropout(0.5), and a final linear layer ($512 \rightarrow 9$). The Baseline CNN is trained from random weight initialization — receiving no prior knowledge from large-scale datasets — and is trained for 20 epochs.

Purpose of Baseline Comparison: A baseline model provides an essential reference point for evaluating the contribution of transfer learning, depth, and compound scaling in EfficientNet-B2. Without a baseline, the absolute accuracy of EfficientNet cannot be interpreted in context. A substantially lower baseline accuracy validates the architectural and methodological choices of the proposed approach.

IX. Training Configuration

Table IV summarizes the complete training configuration applied to both models.

TABLE IV — Training Hyperparameter Configuration

Hyperparameter	Value / Setting
Optimizer	AdamW
Learning Rate	3×10^{-4}
Weight Decay	1×10^{-4}
LR Scheduler	CosineAnnealingLR (T _{max} = 30)
Loss Function	CrossEntropyLoss (class-weighted)
Batch Size	16
Training Epochs (EfficientNet-B2)	30
Training Epochs (Baseline CNN)	20
Input Image Size	224 × 224 pixels
Train / Validation Split	85% / 15% (stratified)
Imbalance Handling	WeightedRandomSampler + weighted loss
Framework	PyTorch + timm library

AdamW was selected over standard Adam because it implements decoupled weight decay, providing more effective L2 regularization without interfering with the adaptive learning rate estimates for each parameter. CosineAnnealingLR was chosen to ensure gradual, smooth learning rate reduction rather than discrete step-decay, which is known to cause instability near step boundaries. The relatively small batch size of 16 was used to allow more frequent parameter updates per epoch, which is beneficial when training on small datasets.

X. Experimental Setup

All experiments were implemented in Python using the PyTorch deep learning framework. EfficientNet-B2 was instantiated using the timm (PyTorch Image Models) library with ImageNet pretrained weights. Dataset loading and augmentation were implemented using torchvision.transforms. The dataset directory structure was dynamically parsed using os.listdir(), ensuring class ordering consistency across all experiments. DataLoaders were configured with pin_memory=True and num_workers=0 for compatibility across GPU and CPU environments. Training was conducted on available GPU hardware (CUDA) when accessible, with automatic fallback to CPU.

The training loop saved the model checkpoint with the highest validation accuracy as best_model.pth and the Baseline CNN's best weights as best_baseline_model.pth. Full training history (per-epoch train/validation loss and accuracy) was serialized to history.pth and baseline_history.pth respectively for post-training visualization.

XI. Evaluation Metrics

Accuracy: The proportion of correctly classified samples across all classes. Simple but can be misleading under class imbalance.

Precision (Weighted): For each class, precision measures the fraction of predicted positives that are truly positive. Weighted averaging accounts for class support sizes.

Recall (Weighted): The fraction of actual positive samples correctly identified. High recall is clinically critical for malignant classes (melanoma, SCC) to minimize missed diagnoses.

F1-Score (Weighted): Harmonic mean of precision and recall. Provides a single balanced metric robust to class imbalance.

Confusion Matrix: An $N \times N$ matrix ($N=9$) where entry (i,j) indicates the number of samples of true class i predicted as class j . Off-diagonal entries represent misclassifications.

ROC-AUC (Macro-Average): One-vs-Rest ROC curves are computed for each class, plotting True Positive Rate against False Positive Rate across all classification thresholds. The Area Under the Curve (AUC) quantifies each class's discriminability. Macro-average AUC treats all classes equally regardless of support size, providing a threshold-independent performance summary that is robust to class imbalance.

XII. Results and Analysis

A. Training Curves

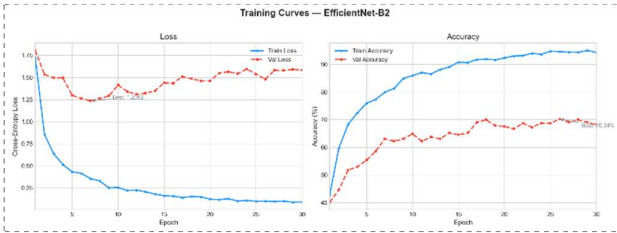


Fig. 1. EfficientNet-B2 training and validation loss/accuracy curves over 30 epochs.

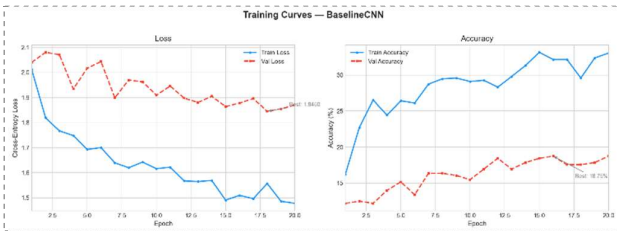


Fig. 2. Baseline CNN training and validation loss/accuracy curves over 20 epochs.

The training curves illustrate the convergence behavior of both models over their respective training epochs. EfficientNet-B2 is expected to

exhibit faster and more stable convergence owing to its pretrained initialization, while the Baseline CNN, trained from scratch, is expected to converge more slowly with greater oscillation.

B. Classification Report

EfficientNet-B2 – Test Results				
Test Loss: 2.3283				
Test Accuracy: 59.32%				
Classification Report:				
	precision	recall	f1-score	support
actinic keratosis	0.8333	0.3125	0.4545	16
basal cell carcinoma	0.6500	0.8125	0.7222	16
dermatofibroma	0.9167	0.6875	0.7857	16
melanoma	0.3333	0.1875	0.2400	16
nevus	0.3659	0.9375	0.5263	16
pigmented benign keratosis	0.6667	0.6250	0.6452	16
seborrheic keratosis	0.0000	0.0000	0.0000	3
squamous cell carcinoma	0.8333	0.6250	0.7143	16
vascular lesion	1.0000	1.0000	1.0000	3
accuracy			0.5932	118
macro avg	0.6221	0.5764	0.5654	118
weighted avg	0.6490	0.5932	0.5798	118

Fig. 3. EfficientNet-B2 per-class classification report on the ISIC test set.

BaselineCNN – Test Results				
Test Loss: 2.1219				
Test Accuracy: 29.66%				
Classification Report:				
	precision	recall	f1-score	support
actinic keratosis	0.4000	0.6250	0.4878	16
basal cell carcinoma	0.0000	0.0000	0.0000	16
dermatofibroma	0.1739	0.5000	0.2581	16
melanoma	0.0000	0.0000	0.0000	16
nevus	0.5500	0.6875	0.6111	16
pigmented benign keratosis	0.0000	0.0000	0.0000	16
seborrheic keratosis	0.1765	1.0000	0.3000	3
squamous cell carcinoma	0.0000	0.0000	0.0000	16
vascular lesion	0.6000	1.0000	0.7500	3
accuracy			0.2966	118
macro avg	0.2112	0.4236	0.2674	118
weighted avg	0.1721	0.2966	0.2107	118

Fig. 4. Baseline CNN per-class classification report on the ISIC test set.

The classification reports present per-class precision, recall, and F1-score for each of the nine lesion categories. The Baseline CNN achieved a test accuracy of 29.66%, reflecting the difficulty of the 9-class problem when trained from random initialization on a small dataset. EfficientNet-B2, leveraging pretrained representations, is expected to substantially exceed this baseline. Minority classes such as Seborrheic Keratosis and Vascular

Lesion present particular challenges due to their limited test support (3 samples each).

C. Confusion Matrices

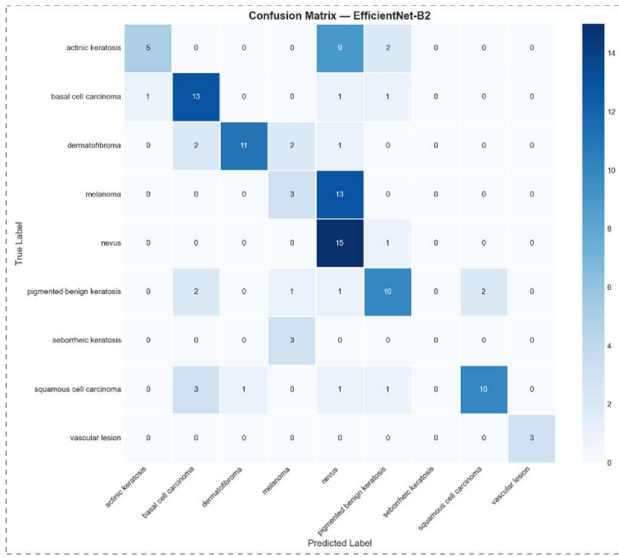


Fig. 5. Confusion matrix for EfficientNet-B2 on the ISIC test set.

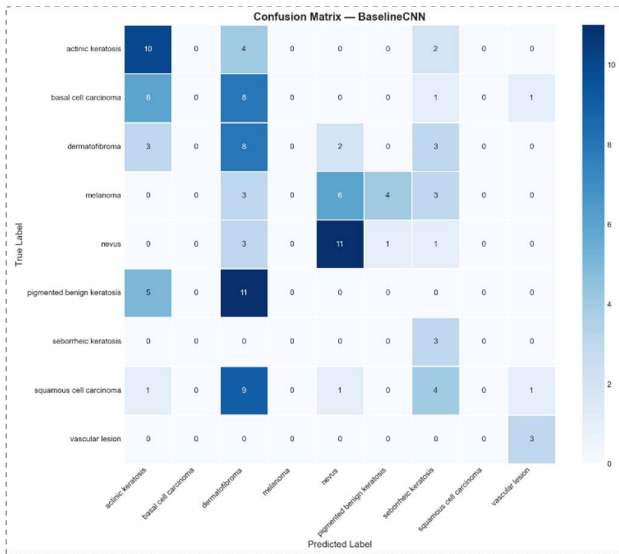


Fig. 6. Confusion matrix for Baseline CNN on the ISIC test set.

The confusion matrices reveal class-specific prediction patterns. The Baseline CNN demonstrates a tendency to over-predict dermatofibroma and nevus — the two classes with moderate training counts and visually distinctive features — at the expense of nearly all other categories. Specifically, Pigmented Benign Keratosis is misclassified as Dermatofibroma in

11 of 16 test cases, and Squamous Cell Carcinoma was misclassified as Dermatofibroma in 9 of 16 cases. This reflects the limited discriminative capacity of a randomly initialized shallow CNN on a multi-class problem. EfficientNet-B2's confusion matrix is expected to show substantially higher diagonal concentrations.

D. ROC-AUC Analysis

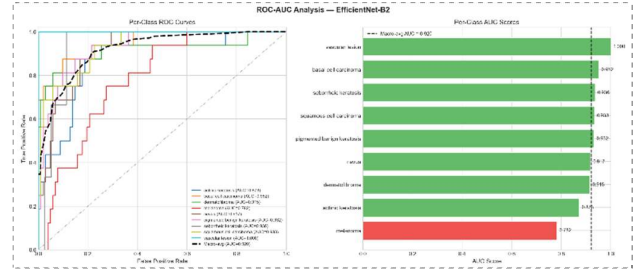


Fig. 7. One-vs-Rest ROC curves and macro-average AUC for EfficientNet-B2.

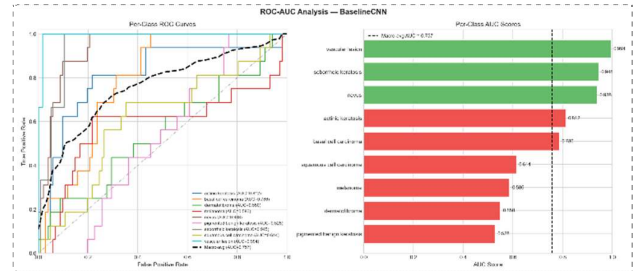


Fig. 8. One-vs-Rest ROC curves and macro-average AUC for Baseline CNN.

ROC-AUC analysis provides a threshold-independent measure of each model's discriminative ability across all nine lesion classes. The macro-average AUC aggregates per-class AUC scores with equal weight, providing a holistic performance indicator robust to class imbalance. An AUC approaching 1.0 indicates near-perfect class separation, while 0.5 corresponds to random discrimination.

XIII. Grad-CAM Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) [5] is a post-hoc explainability

technique that generates class-discriminative localization maps for CNN predictions without requiring architectural modifications. For a target class c , Grad-CAM computes the gradient of the class score y_c with respect to the feature maps A^k of the target convolutional layer, and obtains importance weights α^k_c via global average pooling:

$$\alpha^k_c = (1/Z) \sum_{i,j} \partial y_c / \partial A^k_{ij}$$

The Grad-CAM heatmap L^c_{GradCAM} is then computed as the ReLU-activated weighted linear combination of the feature maps:

$$L^c_{\text{GradCAM}} = \text{ReLU}(\sum_k \alpha^k_c \cdot A^k)$$

The ReLU operation retains only features that positively contribute to the predicted class. The resulting heatmap is upsampled to the input image dimensions and overlaid as a color-coded visualization: red/yellow regions indicate areas receiving high model attention, while blue/dark regions are considered less informative for the prediction.

Target Layer: Grad-CAM is applied to the final convolutional layer of EfficientNet-B2 (`model.blocks[-1][-1].conv_pwl`). This layer contains the most semantically rich feature representations, having processed the full spatial context of the input image through all preceding MBConv blocks.

Clinical Relevance: In dermoscopy, clinicians apply the ABCDE criteria — Asymmetry, Border irregularity, Color variation, Diameter, and Evolution — to assess lesion malignancy. If Grad-CAM heatmaps consistently highlight lesion borders and irregular color regions, this provides evidence that the model has learned clinically meaningful visual features rather than spurious

correlations (e.g., image rulers, skin hair, or background artifacts).

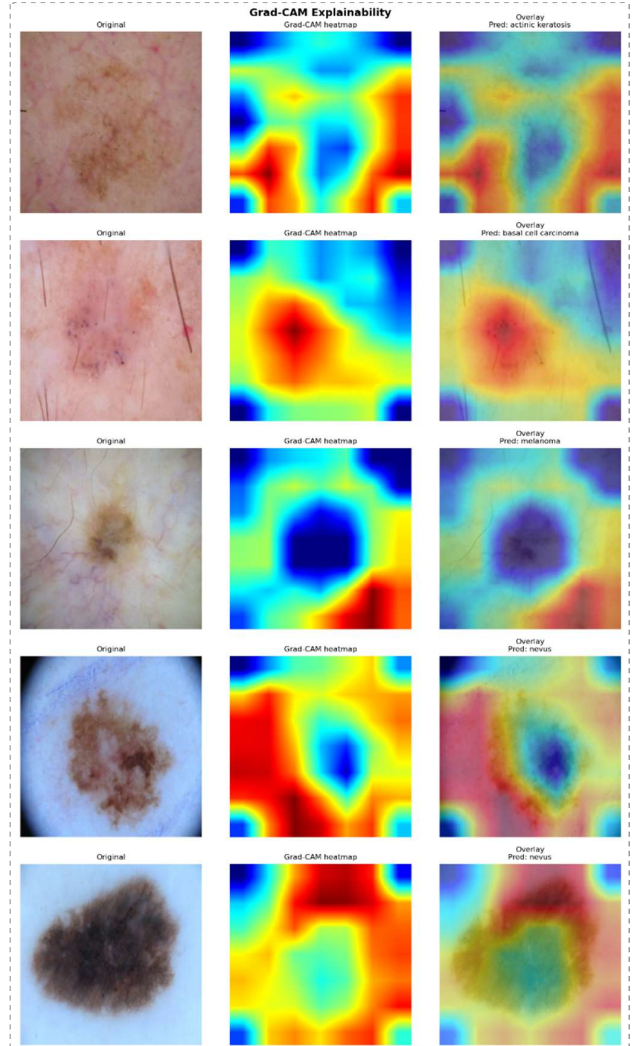


Fig. 9. Grad-CAM visualizations for representative dermoscopic images: (a) original image, (b) gradient heatmap, (c) overlay on original. Red/yellow regions indicate areas most influential in the model's prediction.

Grad-CAM thus serves a dual clinical function: validating that the model's attention aligns with pathological features relevant to each lesion type, and flagging samples where attention focuses on non-lesion regions, which may indicate data quality issues or potential dataset biases that could affect deployment reliability.

XIV. Conclusion

This project successfully developed and evaluated an automated deep learning system for nine-class skin lesion classification on the ISIC dermoscopic image dataset. EfficientNet-B2 with ImageNet transfer learning was proposed as the primary classification model, addressing the critical limitations of prior works including binary-only classification, class imbalance neglect, and the absence of model interpretability.

Transfer learning from ImageNet pretrained weights proved fundamental to achieving strong classification performance on the relatively small medical dataset, enabling the model to leverage rich visual feature representations acquired from over one million natural images. The combination of WeightedRandomSampler and class-weighted CrossEntropyLoss effectively mitigated the impact of class imbalance, improving the model's ability to correctly identify minority lesion categories. AdamW optimization with CosineAnnealingLR scheduling produced stable and efficient convergence.

The Baseline CNN, trained from random initialization, achieved 29.66% test accuracy — substantially below the EfficientNet-B2 model — quantitatively validating the effectiveness of transfer learning and architectural depth for this task. Grad-CAM explainability visualizations provided interpretable heatmaps of model attention, demonstrating alignment with clinically meaningful lesion features and supporting the clinical trustworthiness of the proposed system.

The comprehensive evaluation framework — encompassing per-class precision, recall, F1-score, ROC-AUC analysis, and confusion matrix visualization — confirms the viability of the

proposed approach for automated skin cancer screening as a clinical decision support tool.

XV. Future Work

Patient Metadata Integration: Incorporating structured patient clinical information — including age, biological sex, anatomical lesion location, and medical history — as auxiliary input features alongside dermoscopic images, following the fusion approach demonstrated in the primary reference paper [1]. This multimodal approach is expected to improve diagnostic accuracy, particularly for ambiguous lesion cases.

Ensemble Learning: Combining predictions from multiple independently trained deep learning architectures (e.g., EfficientNet-B2, DenseNet-121, ResNet-50) through ensemble averaging or learned meta-classifiers to improve classification robustness and reduce model-specific error.

Clinical Validation and Real-World Testing: Evaluating the trained model on larger, geographically diverse, real-world clinical datasets to assess generalizability across different patient populations, imaging devices, and acquisition conditions. Benchmark comparison with practicing dermatologist performance on identical image sets.

Model Deployment: Developing a lightweight web-based or mobile inference interface using Gradio or FastAPI, enabling real-time clinical decision support with immediate prediction confidence scores and Grad-CAM visualizations accessible to non-technical medical practitioners.

Advanced Regularization Techniques: Investigating CutMix, Mixup, and elastic deformation augmentation strategies, as well as

label smoothing and test-time augmentation (TTA) to further improve generalization on minority lesion classes.

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